

- (c) Construct LR (0) parsing table for the following grammar :

$$S \rightarrow cB \mid ccA$$

$$A \rightarrow cA \mid a$$

$$B \rightarrow ccB \mid b$$

5. (a) What is three address code? Explain various three address code statement used in the representation.
- (b) What is Backpatching? Explain backpatching scheme for Boolean expression.
- (c) Explain quadruple, triple, indirect triple with suitable example.
6. (a) Give various method of loop optimization.
- (b) What is DAG in code optimization? Construct DAG for the given code by generating three address code :

$$P = Q * - C + (a + b)^* + (a + b)^* D$$

- (c) Explain Peephole optimization.

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S.No. : 292

BCS 3604

No. of Printed Pages : 04

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 33228

Roll
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B. Tech. Examination 2023-24

(Even Semester)

COMPILER DESIGN

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : 8×1=8
- (a) What is two pass compiler?
- (b) Define token and lexeme.
- (c) Give the limitation of top-down parser.
- (d) List the method of error recovery.
- (e) What is intermediate code?
- (f) What is syntax tree?

- (g) Explain the need of code optimization.
 (h) Define leader in basic block.

SECTION - B

2. Attempt any two parts of the following : $2 \times 6 = 12$

- (a) Explain the six phase model of compiler design with example.
 (b) Consider the following grammar find the FIRST and FOLLOW of the non-terminal and check whether it is LL (1) or not :

$$S \rightarrow ACB \mid CbB \mid Ba$$

$$A \rightarrow da \mid BC$$

$$B \rightarrow g \mid \epsilon$$

$$C \rightarrow h \mid c$$

- (c) Generate three address code for the following program segment :

While ($a < c$ or $b > d$) do

If $a = 1$ then

$c = a * b + c$

else

$a = b - d * c$

- (d) Write different method of optimization techniques, explain with suitable example.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Write short notes on the following :
 (i) Role of lexical analyzer
 (ii) Bootstrapping
 (b) What is top-down parser? Explain the working of a predictive parser.
 (c) Design the DFA for $(a \mid b)^* a$ using null moves and minimize it.
4. (a) Consider the following grammar and remove the left recursion and left factor from the grammar :

$$S \rightarrow Aa \mid b$$

$$A \rightarrow aAB \mid aA \mid Ac \mid Sd \mid a \mid \epsilon$$

$$B \rightarrow c \mid \epsilon$$

- (b) What are the different methods of organizing the symbol table? Explain in detail.

/ P. T. O.